

CCNA Practical Lab Report

Connecting Two Computers on a LAN: IP Configuration, Ping, RDP & File Sharing

Lab Environment	QEMU Virtual Machines (Windows 7)
Systems Used	System 1 (Win1) System 2 (Win2)
IP Addressing	70.0.0.40 / 8 (System 1) and 70.0.0.50 / 8 (System 2)
Topics Covered	Static IP Configuration, IPv4 Ping Test, Computer Name & Workgroup, Name-based Ping Test, Remote Desktop (RDP) Configuration

Objective

The objective of this lab is to connect two virtual Windows 7 machines (System 1 and System 2) over a simulated LAN, configure static IPv4 addressing, verify Layer-3 connectivity using **ping**, configure host names for name-based connectivity, and enable **Remote Desktop (RDP)** access along with file sharing between the two hosts.

Part 1 — Static IP Address Configuration

The Network Connections window was opened on both machines using **Run** → **ncpa.cpl**, and a static IPv4 address was assigned to the Local Area Connection adapter on each system.

Step 1.1 — System 1: Opening Network Connections

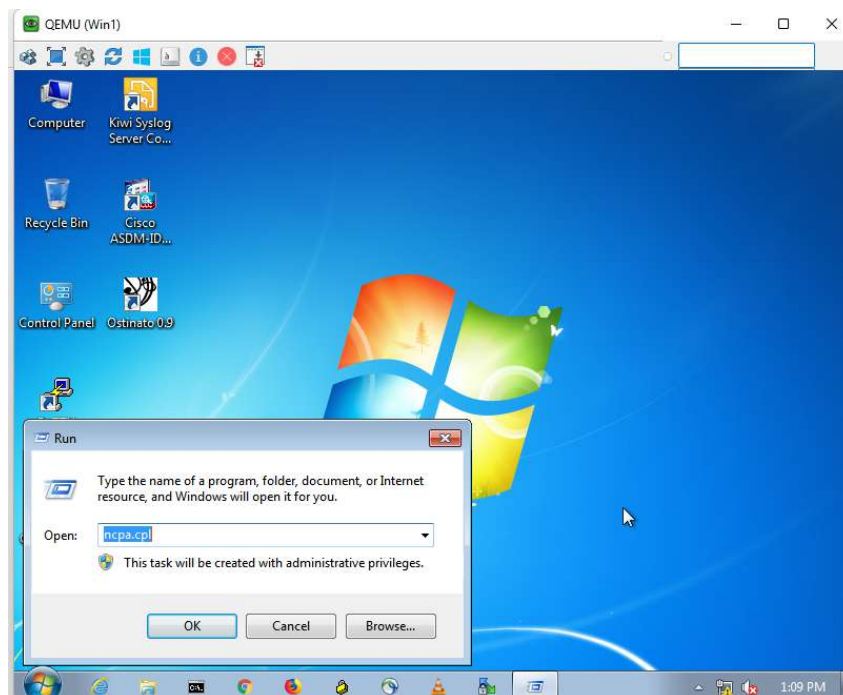


Fig 1.1 – Running **ncpa.cpl** from the Run dialog on System 1.

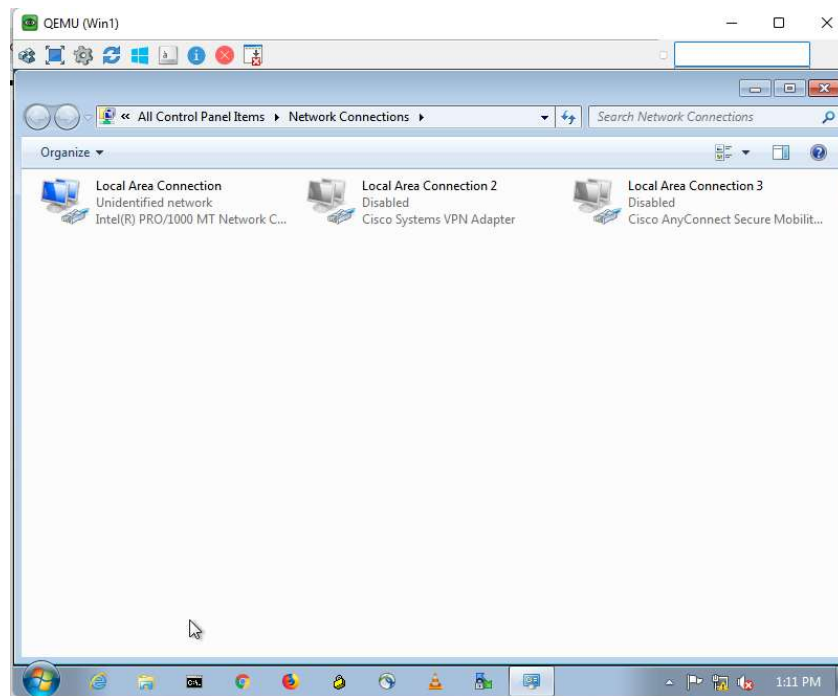


Fig 1.2 – Network Connections window on System 1, showing the Local Area Connection adapter.

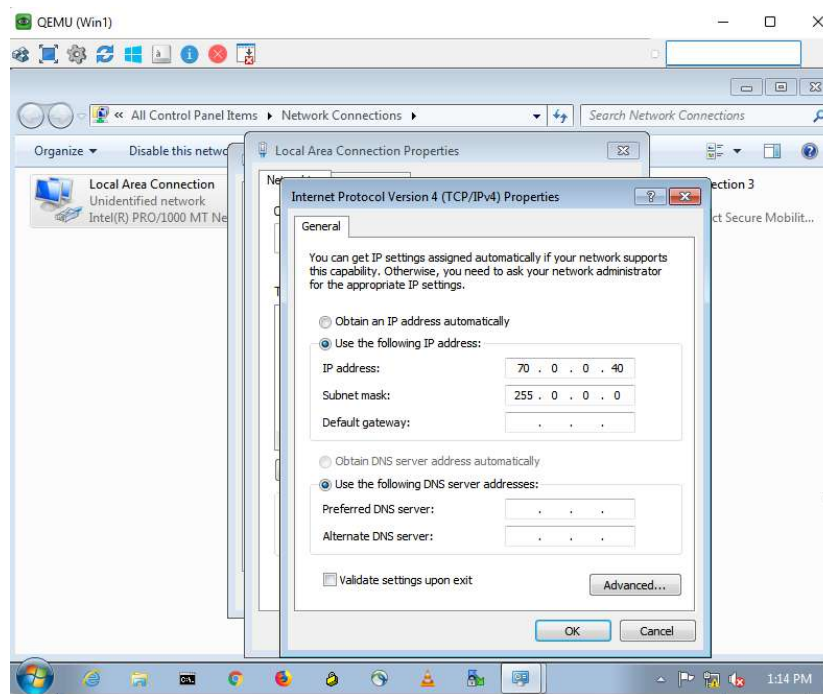


Fig 1.3 – Assigning static IPv4 address **70.0.0.40** with subnet mask **255.0.0.0** on System 1.

Step 1.2 — System 2: Opening Network Connections

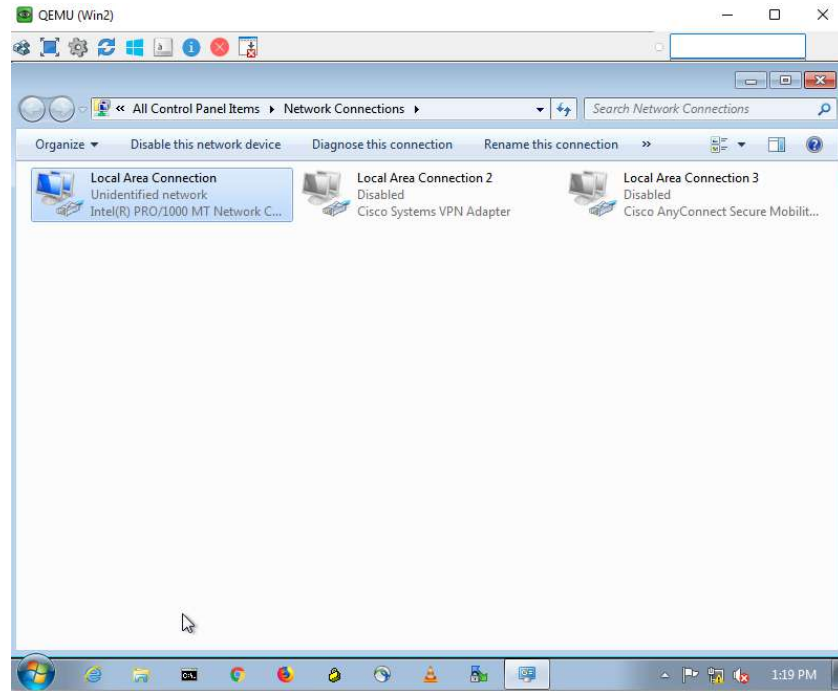


Fig 1.4 – Network Connections window on System 2.

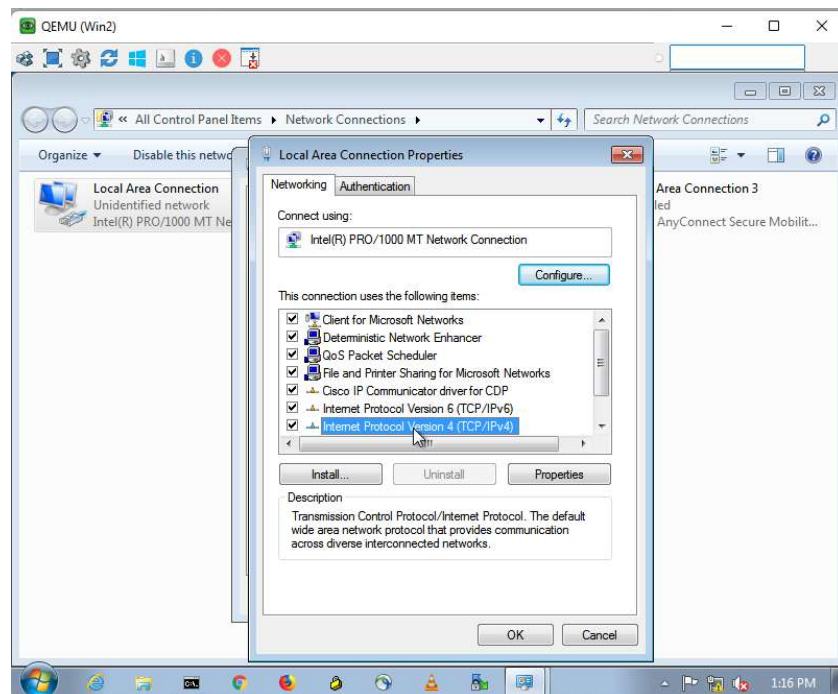


Fig 1.5 – Local Area Connection Properties on System 2, selecting Internet Protocol Version 4 (TCP/IPv4).

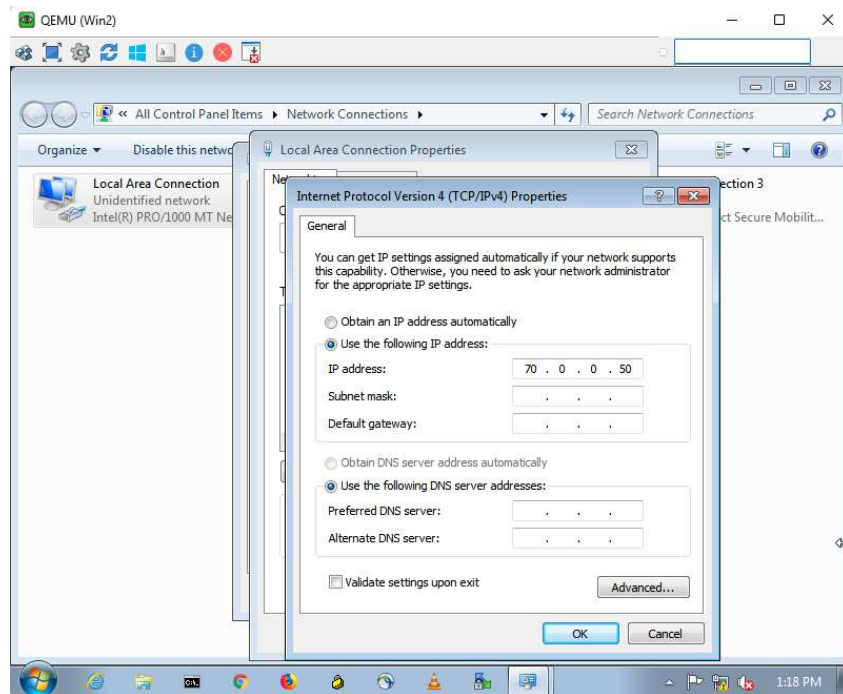


Fig 1.6 – Entering static IP address **70.0.0.50** on System 2 (subnet mask not yet entered).

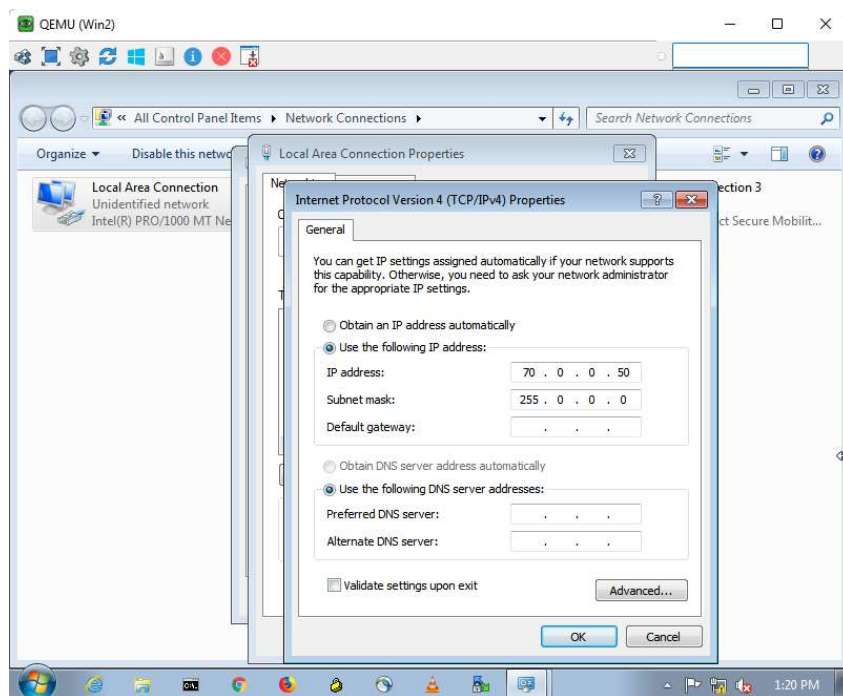
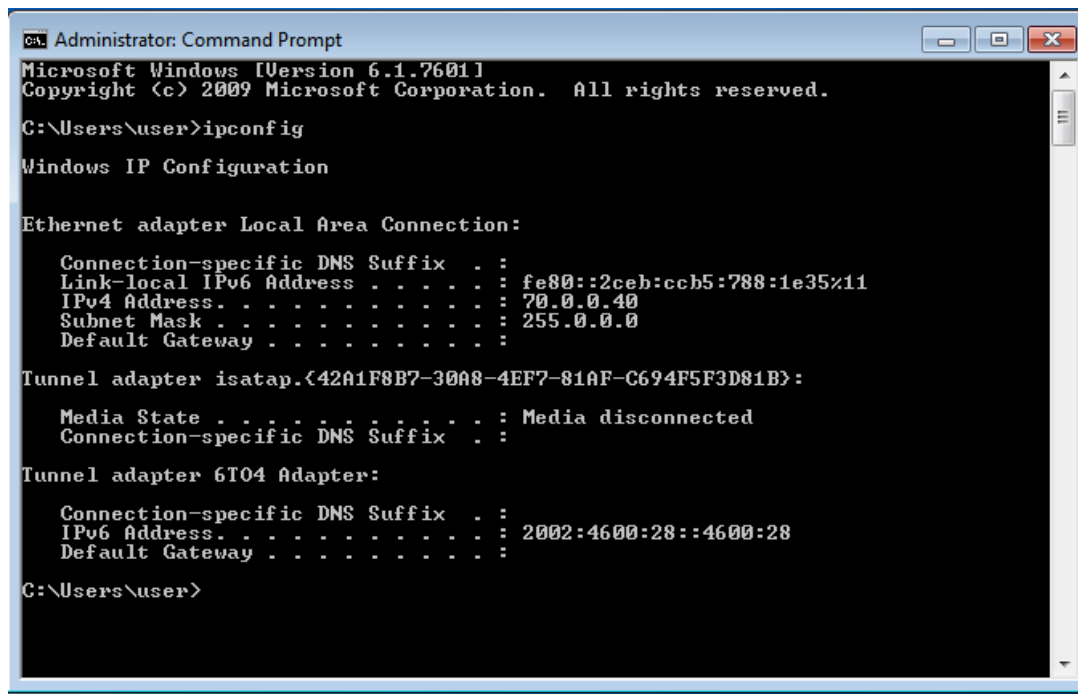


Fig 1.7 – Completed IPv4 configuration on System 2 — IP **70.0.0.50**, subnet mask **255.0.0.0**.

Step 1.3 — Verifying IP Configuration (ipconfig)

The **ipconfig** command was run in an elevated Command Prompt on each system to confirm the assigned IPv4 addresses.



```
Administrator: Command Prompt
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::2ceb:ccb5:788:1e35%11
    IPv4 Address. . . . . : 70.0.0.40
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 

Tunnel adapter isatap.{42A1F8B7-30A8-4EF7-81AF-C694F5F3D81B}:

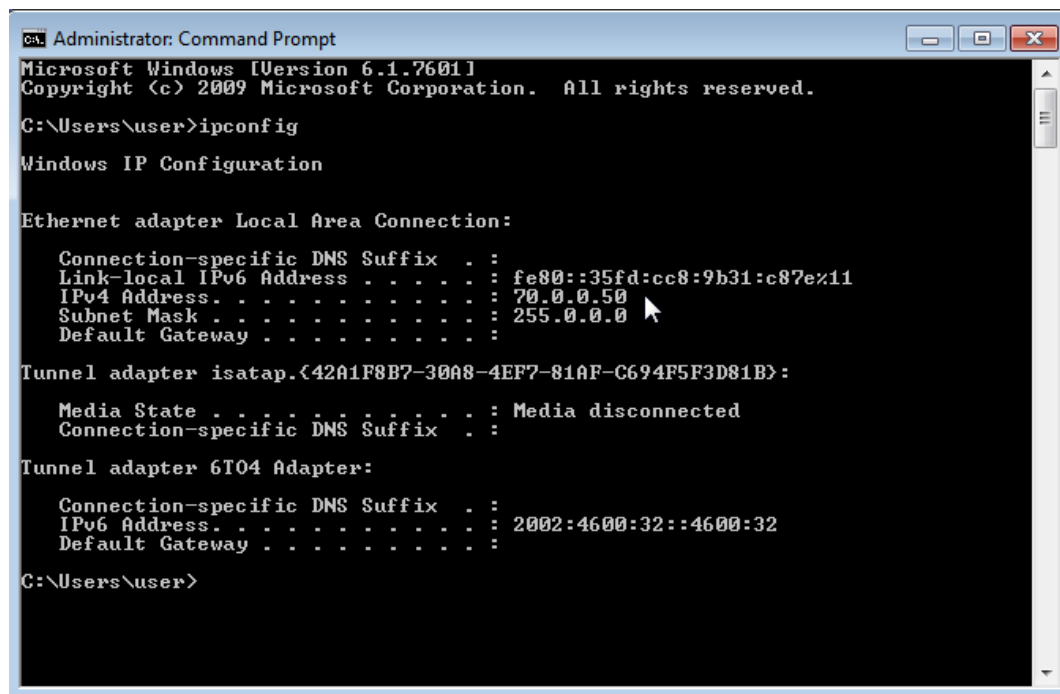
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter 6T04 Adapter:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2002:4600:28::4600:28
    Default Gateway . . . . . : 

C:\Users\user>
```

Fig 1.8 – **ipconfig** output on System 1 confirming IPv4 Address 70.0.0.40, Subnet Mask 255.0.0.0.



```
Administrator: Command Prompt
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::35fd:cc8:9b31:c87e%11
    IPv4 Address. . . . . : 70.0.0.50
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 

Tunnel adapter isatap.{42A1F8B7-30A8-4EF7-81AF-C694F5F3D81B}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter 6T04 Adapter:

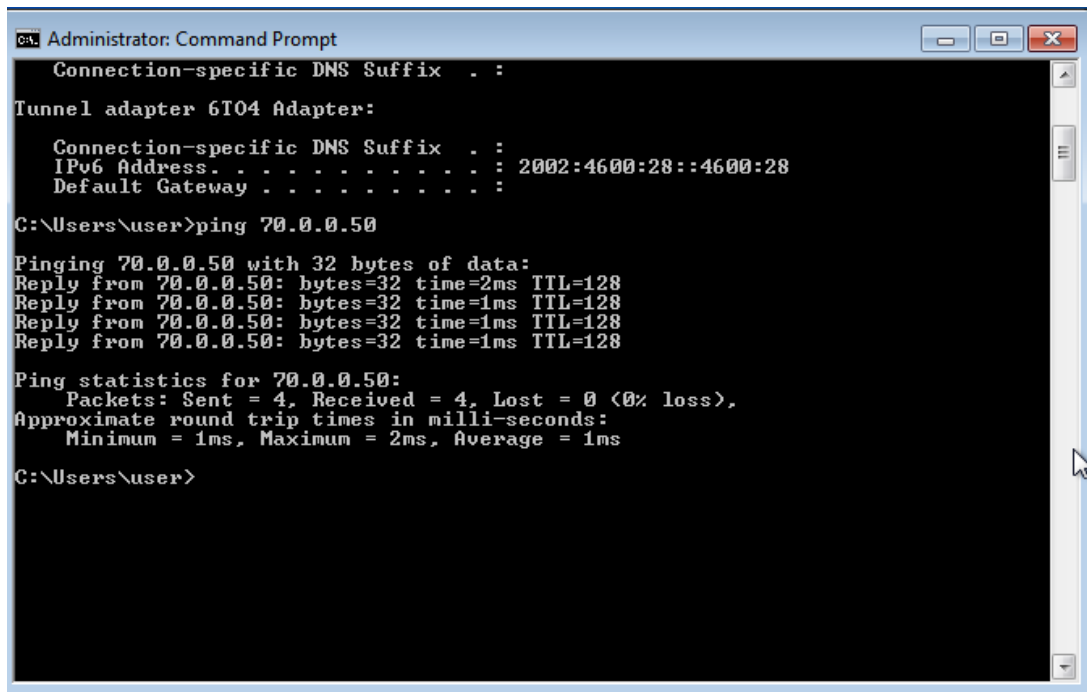
    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2002:4600:32::4600:32
    Default Gateway . . . . . : 

C:\Users\user>
```

Fig 1.9 – **ipconfig** output on System 2 confirming IPv4 Address 70.0.0.50, Subnet Mask 255.0.0.0.

Part 2 — Connectivity Test using IP Address (Ping)

With static IP addresses configured on both machines, the **ping** command was used to verify end-to-end Layer-3 connectivity between System 1 and System 2.



```
Administrator: Command Prompt
Connection-specific DNS Suffix . :
Tunnel adapter 6T04 Adapter:
    Connection-specific DNS Suffix . :
    IPv6 Address. . . . . : 2002:4600:28::4600:28
    Default Gateway . . . . . :

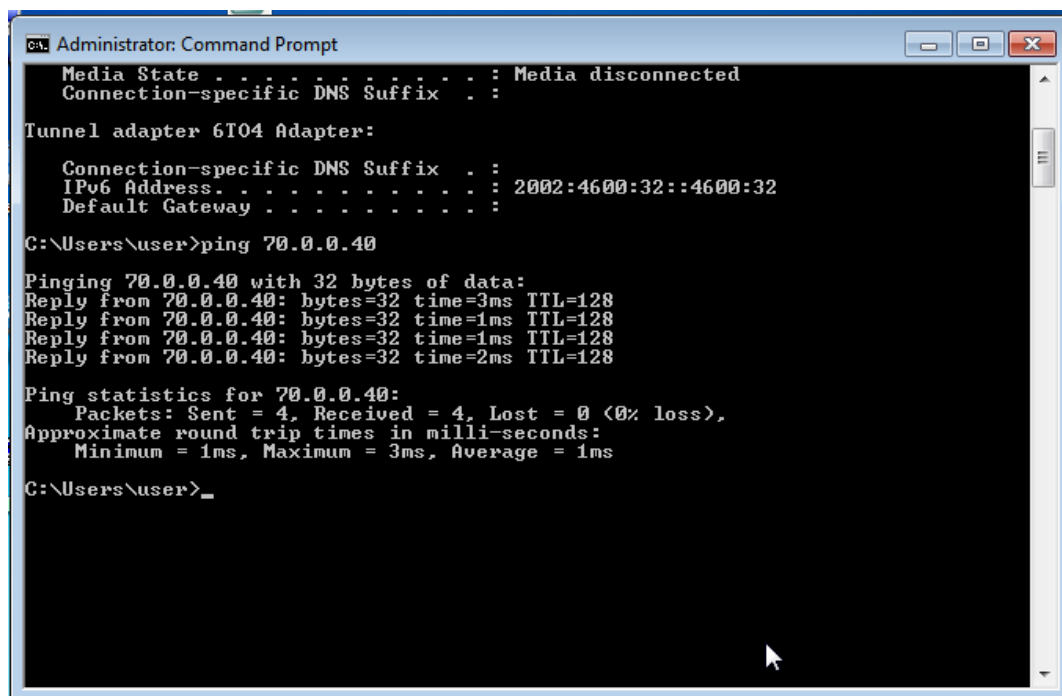
C:\Users\user>ping 70.0.0.50

Pinging 70.0.0.50 with 32 bytes of data:
Reply from 70.0.0.50: bytes=32 time=2ms TTL=128
Reply from 70.0.0.50: bytes=32 time=1ms TTL=128
Reply from 70.0.0.50: bytes=32 time=1ms TTL=128
Reply from 70.0.0.50: bytes=32 time=1ms TTL=128

Ping statistics for 70.0.0.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\user>
```

Fig 2.1 – Pinging 70.0.0.50 (System 2) from System 1 — 4 packets sent, 4 received, 0% loss.



```
Administrator: Command Prompt
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Tunnel adapter 6T04 Adapter:
    Connection-specific DNS Suffix . :
    IPv6 Address. . . . . : 2002:4600:32::4600:32
    Default Gateway . . . . . :

C:\Users\user>ping 70.0.0.40

Pinging 70.0.0.40 with 32 bytes of data:
Reply from 70.0.0.40: bytes=32 time=3ms TTL=128
Reply from 70.0.0.40: bytes=32 time=1ms TTL=128
Reply from 70.0.0.40: bytes=32 time=1ms TTL=128
Reply from 70.0.0.40: bytes=32 time=2ms TTL=128

Ping statistics for 70.0.0.40:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Users\user>
```

Fig 2.2 – Pinging 70.0.0.40 (System 1) from System 2 — 4 packets sent, 4 received, 0% loss.

Result: Both systems successfully reached each other with 0% packet loss, confirming that the LAN is correctly configured at Layer 3.

Part 3 — Computer Name & Workgroup Configuration

To enable name-based access on the LAN (instead of using raw IP addresses), each system's computer name was set/verified via **Run** → **sysdm.cpl** → **Computer Name**. System 1 was renamed to **System 1** and System 2 was renamed to **System2**, with both systems kept in the same **WORKGROUP**.

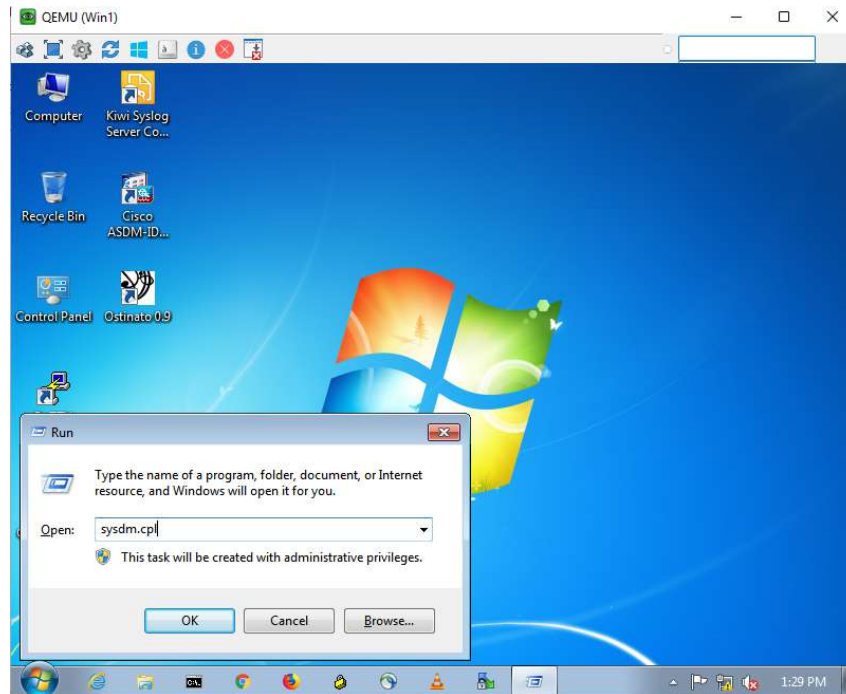


Fig 3.1 – Running **sysdm.cpl** from the Run dialog on System 1.

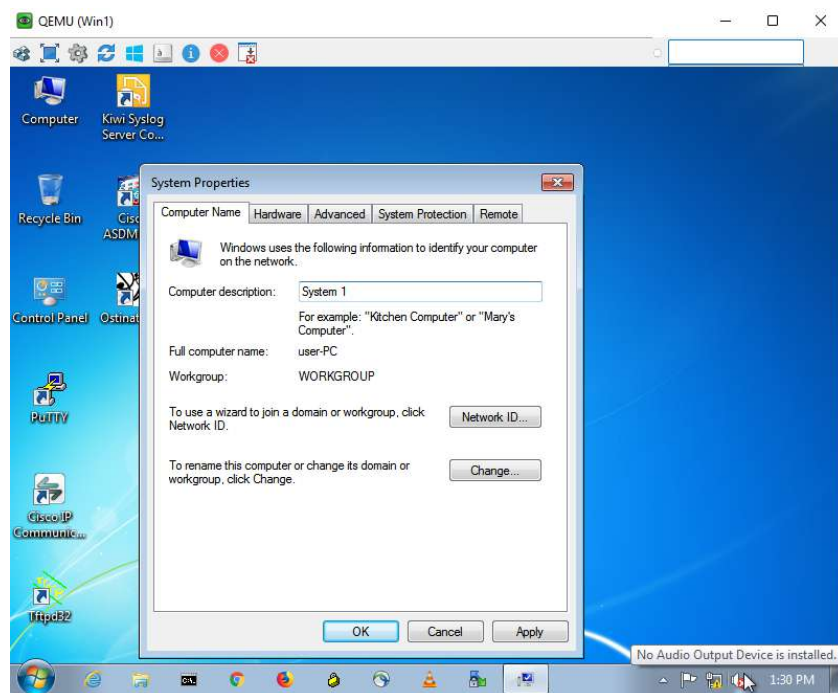


Fig 3.2 – System Properties — Computer Name tab showing full computer name **user-PC** and Workgroup **WORKGROUP**.

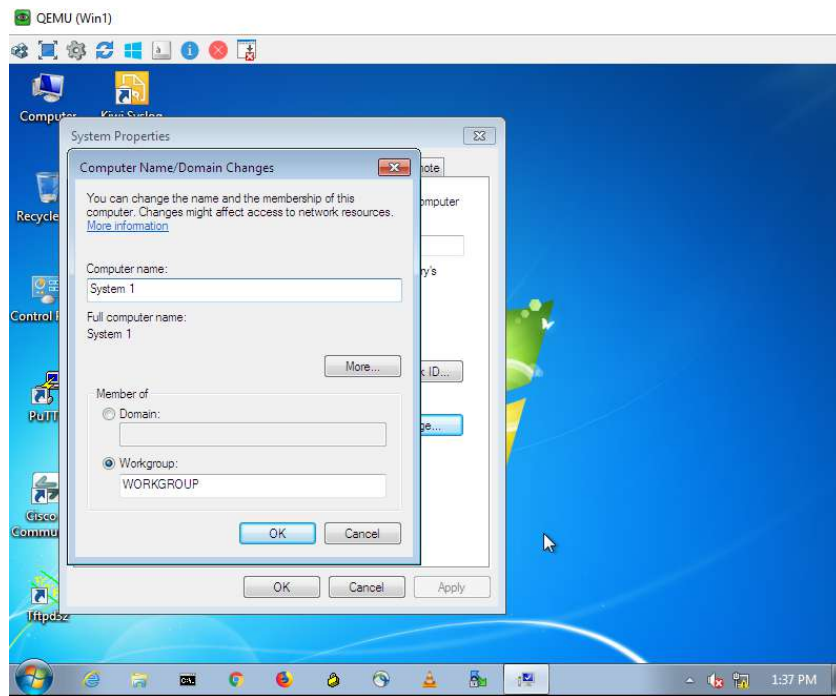


Fig 3.3 – Computer Name/Domain Changes dialog — renaming the computer to **System 1**, member of Workgroup **WORKGROUP**.

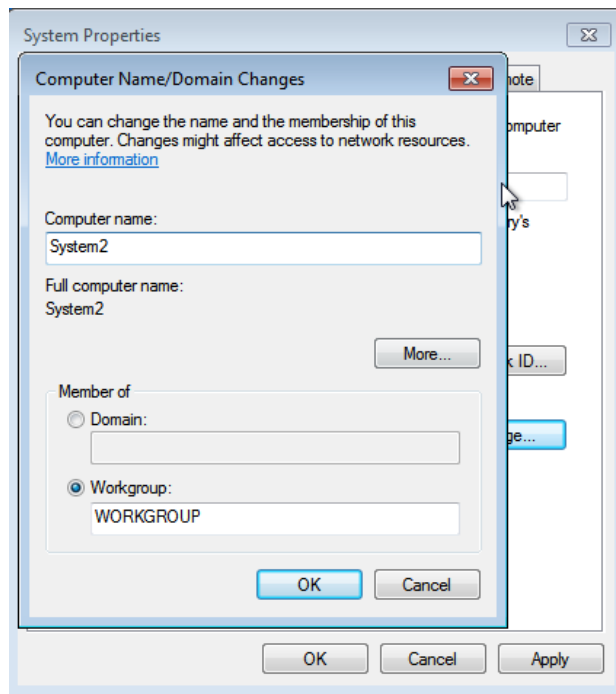
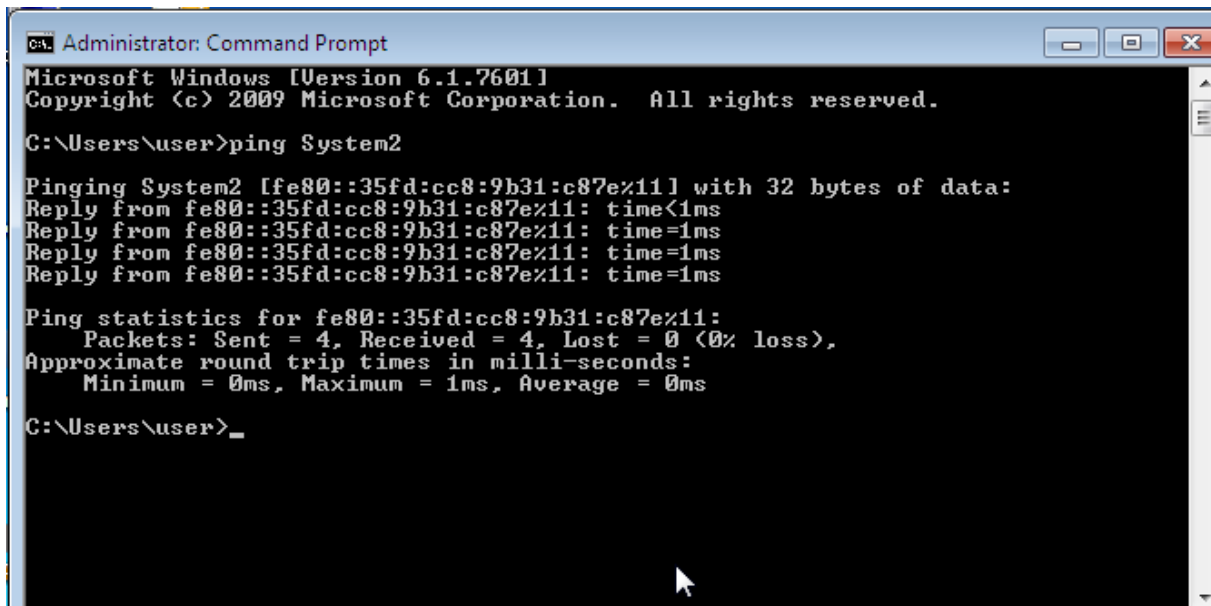


Fig 3.4 – Computer Name/Domain Changes dialog on System 2 — renaming the computer to **System2**, member of Workgroup **WORKGROUP**.

Part 4 — Connectivity Test using Computer Name

After renaming and rebooting/refreshing network settings, connectivity between the two systems was re-verified using the assigned computer names instead of IP addresses.



```
C:\Users\user>ping System2

Pinging System2 [fe80::35fd:cc8:9b31:c87e%11] with 32 bytes of data:
Reply from fe80::35fd:cc8:9b31:c87e%11: time<1ms
Reply from fe80::35fd:cc8:9b31:c87e%11: time=1ms
Reply from fe80::35fd:cc8:9b31:c87e%11: time=1ms
Reply from fe80::35fd:cc8:9b31:c87e%11: time=1ms

Ping statistics for fe80::35fd:cc8:9b31:c87e%11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\user>
```

Fig 4.1 – Pinging **System2** by name from System 1 — resolved to link-local IPv6 address, 4/4 replies received.

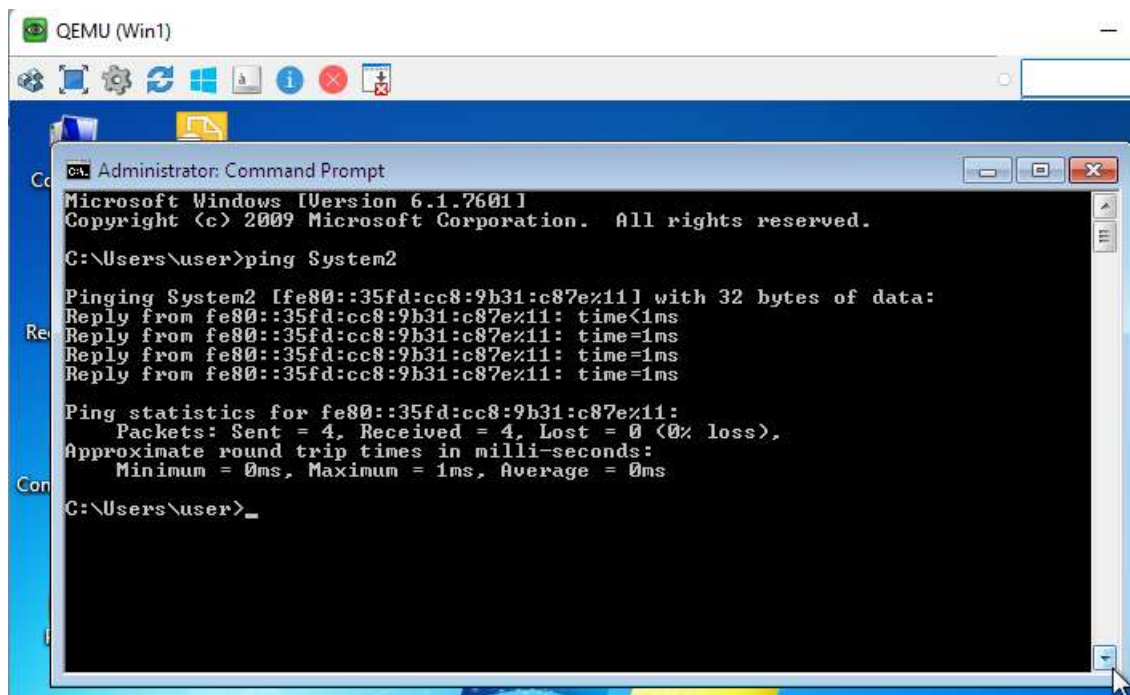


Fig 4.2 – Command Prompt on System 1 (QEMU Win1) showing successful **ping System2** results.

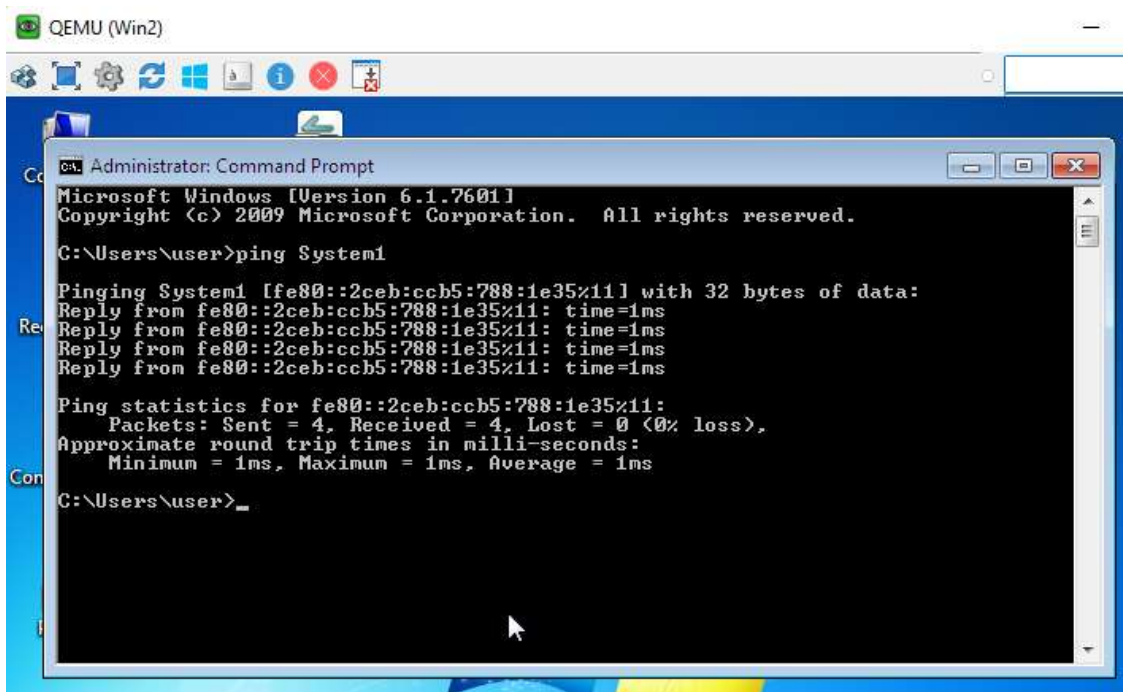


Fig 4.3 – Command Prompt on System 2 (QEMU Win2) showing successful **ping System1** results.

Result: Both systems resolved each other's computer names and responded successfully, confirming NetBIOS/name resolution is working correctly on the LAN.

Part 5 — Enabling Remote Desktop (RDP) & File Sharing

Finally, Remote Desktop access was enabled on System 1 via **System Properties** → **Remote** tab so that it could be remotely accessed and used for file sharing from System 2 over the LAN.



Fig 5.1 – Reopening **sysdm.cpl** on System 1 to access Remote settings.

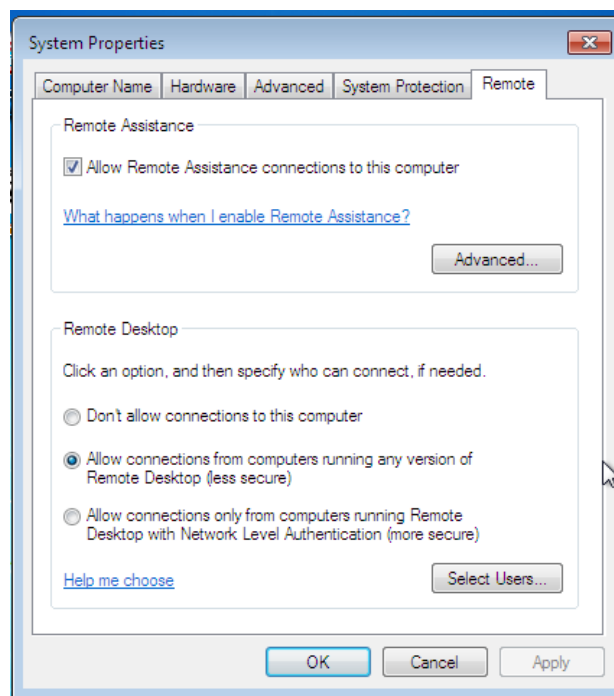


Fig 5.2 – Remote tab — Remote Assistance enabled, and Remote Desktop set to **"Allow connections from computers running any version of Remote Desktop"**.

Result: With Remote Desktop enabled and the two systems able to resolve each other by IP address and computer name, System 2 can now connect to System 1 via RDP and access shared folders/files between the two hosts over the LAN.

Part 6 — Establishing a Remote Desktop Connection

With Remote Desktop enabled on System 1 (Part 5), a live RDP session was initiated using the **mstsc** (Remote Desktop Connection) utility to connect from System 1 to **System2** over the LAN.

Step 6.1 — Launching Remote Desktop Connection (mstsc)

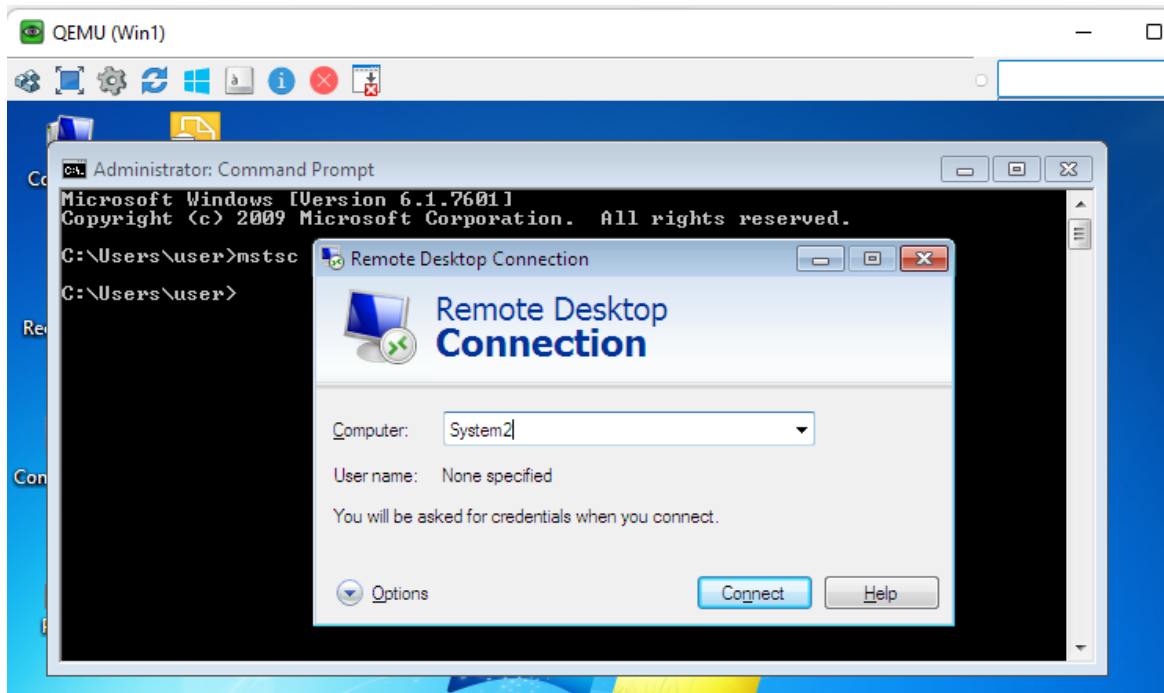


Fig 6.1 – Running **mstsc** from the command prompt and entering the target computer name **System2** in the Remote Desktop Connection dialog.

Step 6.2 — Authenticating the Remote Session

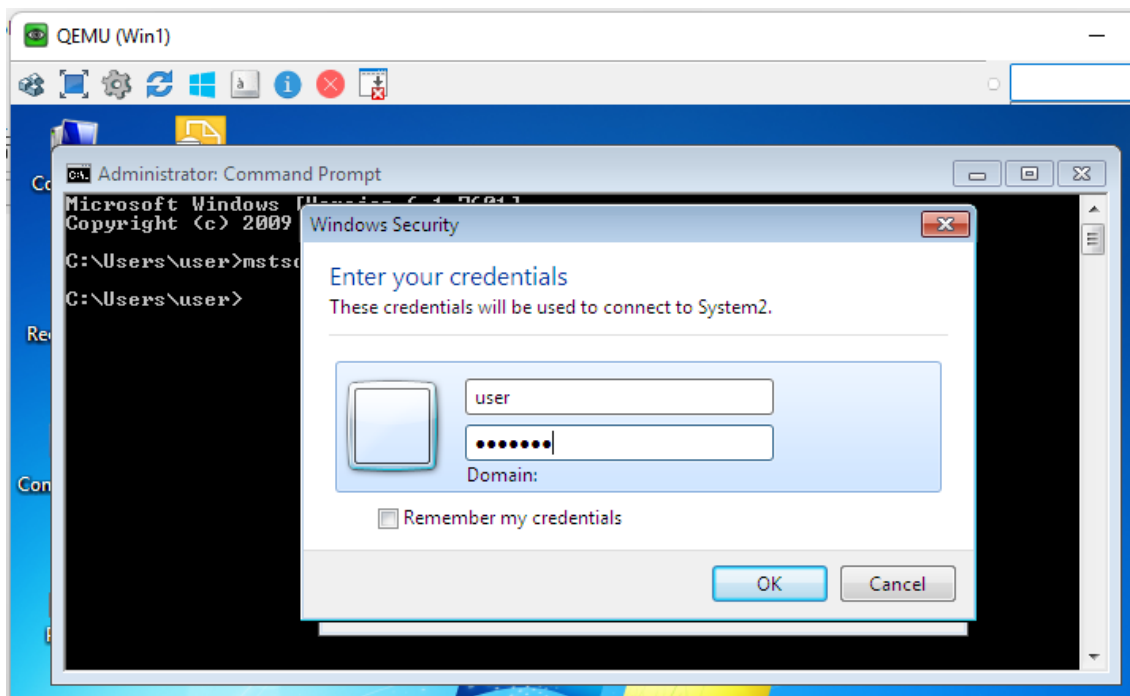


Fig 6.2 – Windows Security prompt requesting credentials to connect to **System2** — username **user** and password entered.

Step 6.3 — Active Remote Desktop Session

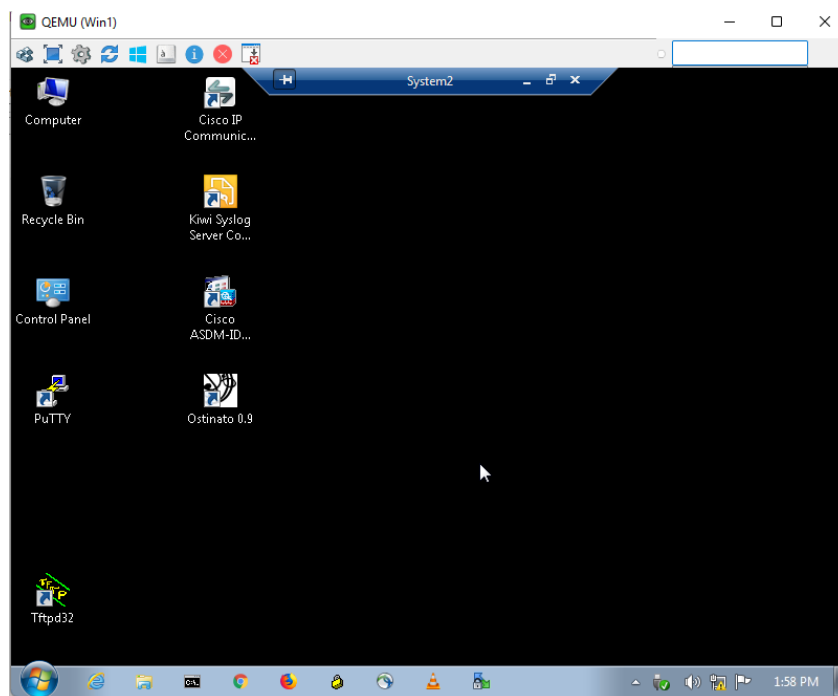


Fig 6.3 – Successful RDP session — System 1 is now remotely controlling the **System2** desktop, shown in an embedded RDP window.

Result: The RDP connection was established successfully, confirming that Remote Desktop is fully functional between the two systems over the LAN.

Part 7 — Configuring Network File Sharing

A folder named **Transfer file** was created on System 1's desktop and shared on the network so that it could be accessed from System 2, demonstrating file-sharing configuration between the two hosts.

Step 7.1 — Creating the Shared Folder

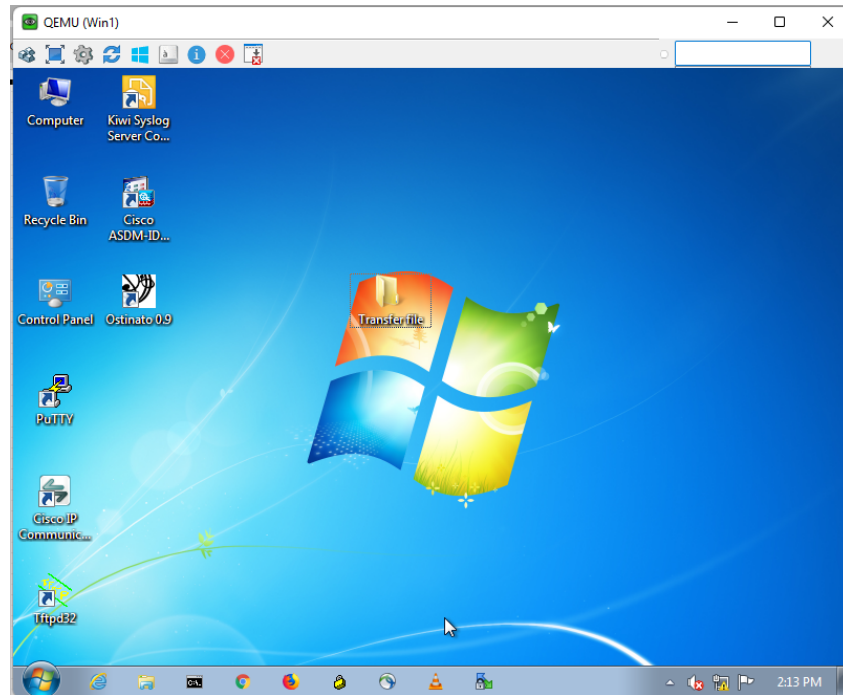


Fig 7.1 – A new folder named **Transfer file** created on the System 1 desktop.

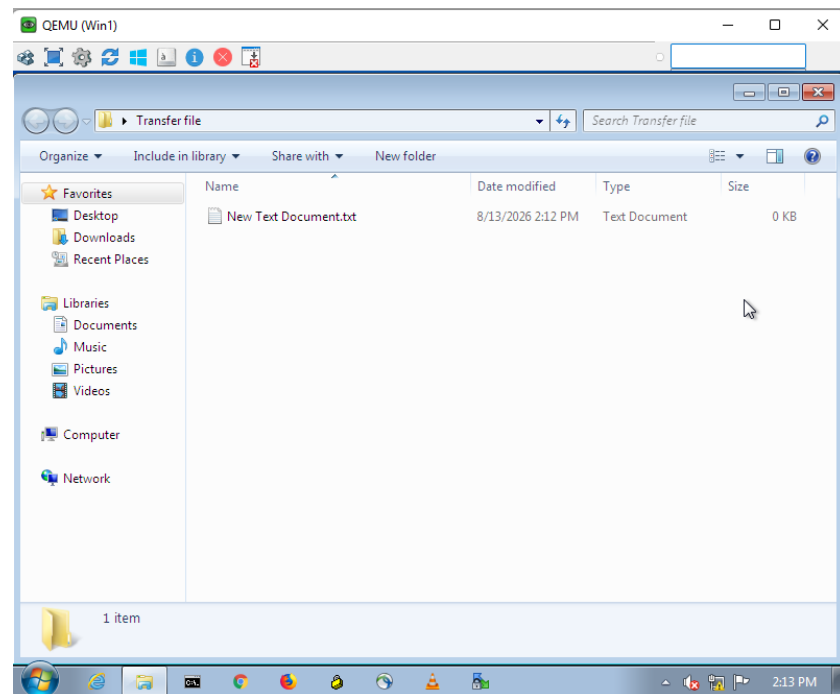


Fig 7.2 – Contents of the **Transfer file** folder, containing **New Text Document.txt** to be shared.

Step 7.2 — Enabling Sharing on the Folder

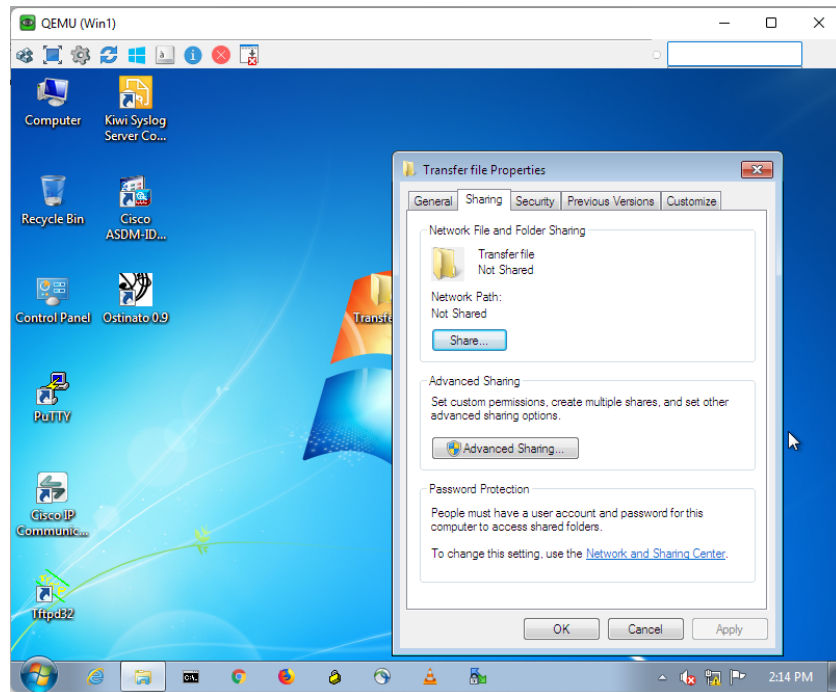


Fig 7.3 – **Transfer file Properties** → **Sharing** tab, showing the folder's current status as **Not Shared** before configuration.

Step 7.3 — Setting Share Permissions

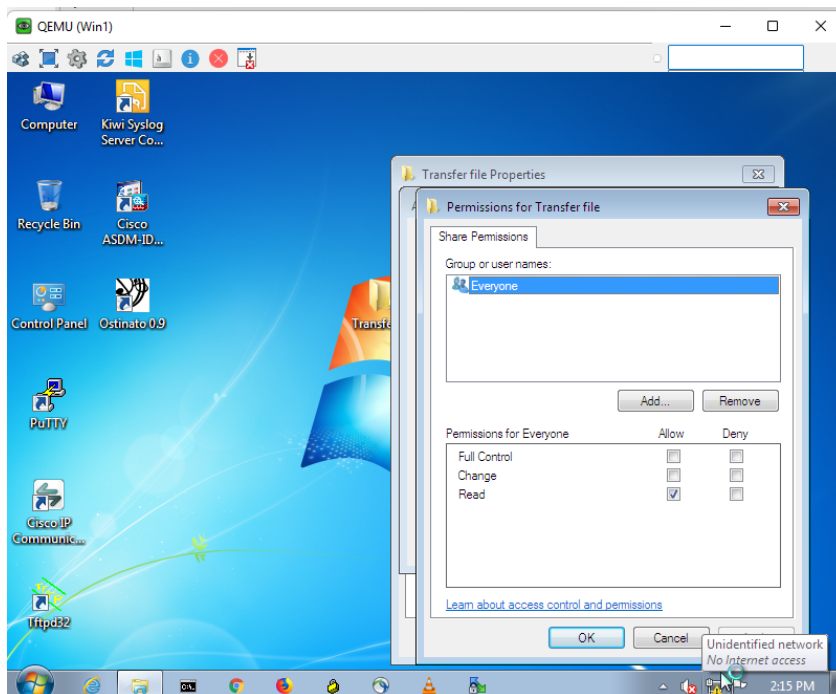


Fig 7.4 – **Share Permissions** dialog for **Transfer file** — the **Everyone** group is granted **Read = Allow** access.

Result: The **Transfer file** folder on System 1 was successfully shared on the network with Read access for all users.

Part 8 — Verifying File Sharing from the Remote System

From System 2, the network was browsed to confirm that System 1 and its shared folder were visible and accessible over the LAN.

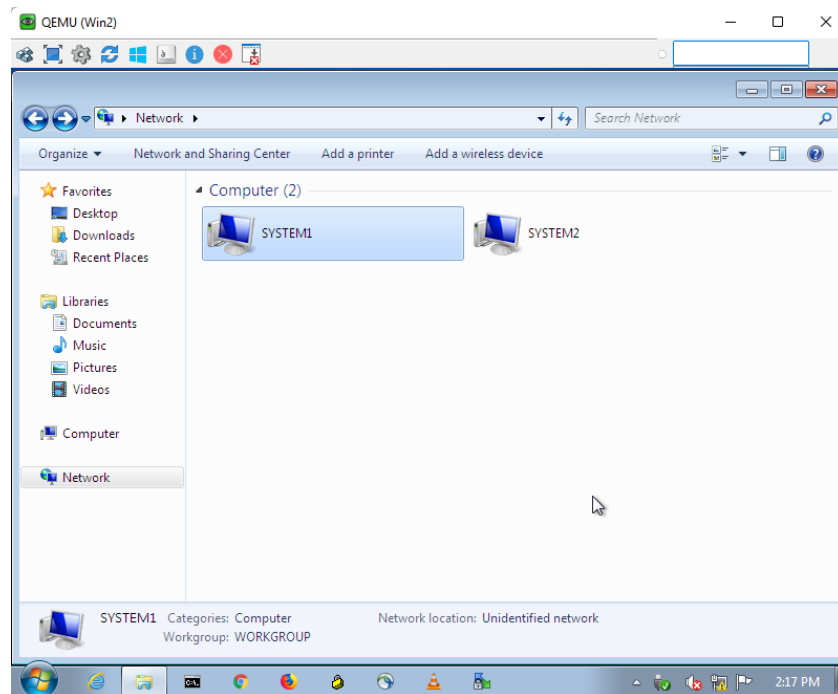


Fig 8.1 – Network view on System 2 showing both **SYSTEM1** and **SYSTEM2** listed as computers on the LAN.

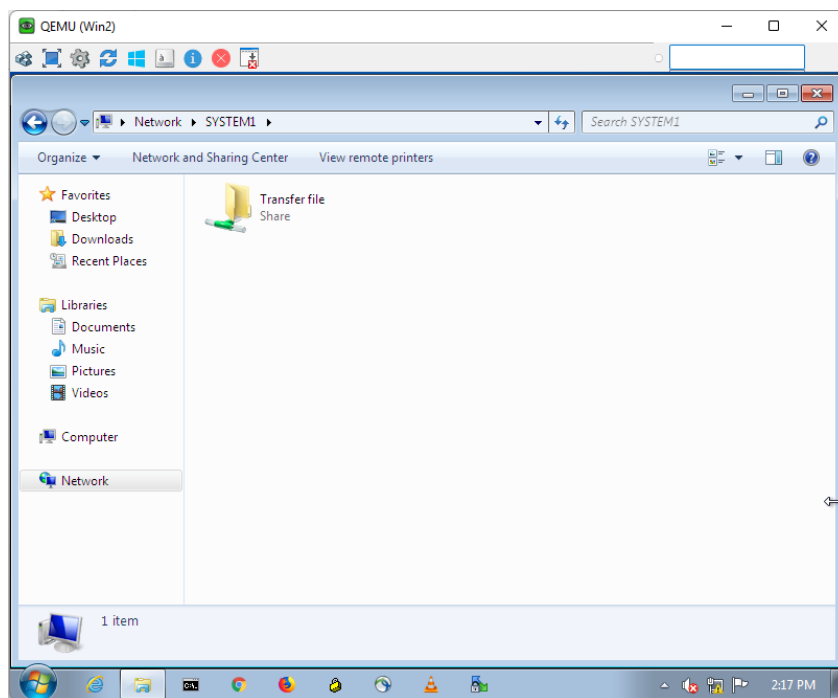


Fig 8.2 – Browsing into **\\SYSTEM1** from System 2, showing the shared **Transfer file** folder available for access.

Result: System 2 successfully discovered System 1 on the network and accessed the shared **Transfer file** folder, confirming that network file sharing is fully functional between the two systems.

Conclusion

In this lab, two Windows 7 virtual machines were successfully connected over a LAN. Static IPv4 addresses (70.0.0.40 and 70.0.0.50, subnet 255.0.0.0) were configured and verified with **ipconfig**. Layer-3 connectivity was confirmed using **ping** by both IP address and computer name (System1 / System2) within a common WORKGROUP. Remote Desktop was enabled on System 1 and a live RDP session was successfully established from System 1 to System2 using **mstsc**. Finally, a folder was shared on System 1 with configured permissions and verified as accessible from System 2 over the network — completing all required objectives of the lab: **IP addressing, ping testing, Remote Desktop Protocol (RDP), and network file sharing**.